

Billboard Management System — Architecture & Design

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1. High-Level Architecture

```
graph TB
    subgraph "Frontend (React + Vite)"
        CP["Client Portal"]
        AP["Admin Dashboard"]
        VP["Vendor Portal"]
    end

    subgraph "Real-Time Layer"
        SIO["Socket.IO Server"]
    end

    subgraph "Backend (Node.js + Express)"
        AUTH["Auth Service<br/>(JWT + Role-based)"]
        CAMP["Campaign Controller"]
        SURV["Site Survey Controller"]
        CREAT["Creatives Controller"]
        WO["Work Order Controller"]
        INST["Installation Controller"]
        INV["Invoice Controller"]
        NOTIF["Notification Controller"]
        PDF["PDF Generator"]
    end

    subgraph "Storage"
        MONGO["MongoDB<br/>billboard_management"]
        UPLOAD["File Storage<br/>(Uploads / Cloud)"]
    end

    subgraph "External Services"
        MAPS["Google Maps API<br/>(GPS & Location)"]
        CAM["Device Camera API"]
    end

    CP & AP & VP -->|"HTTP + WebSocket"| AUTH
```

```
AUTH --> CAMP & SURV & CREAT & WO & INST & INV
CAMP & SURV & CREAT & WO & INST & INV --> MONGO
SURV & INST --> UPLOAD
SURV --> MAPS
SURV --> CAM
NOTIF --> SIO
SIO -->|"Push Notifications"| CP & AP & VP
SURV & INST --> PDF
```

2. Roles

Role	Responsibilities
Client	Creates campaigns, reviews installation reports, approves/rejects invoices
Admin	Reviews campaigns, allocates vendors, selects billboard images, creates work orders, manages creatives, forwards reports/invoices
Vendor	Conducts site surveys (uploads geotagged billboard photos), performs installations, uploads installation images, sends invoices

3. Complete Callflow (6 Phases)

```
sequenceDiagram
    autonumber
    participant C as Client
    participant A as Admin
    participant V as Vendor

    rect rgb(230, 245, 255)
    Note over C,A: PHASE 1 – Campaign Creation & Review
    C->>A: Create Campaign (brief, budget, locations, dates)
    A->>A: Review Campaign Brief
    alt Approved
        A-->>C: ✅ Notify: Campaign Approved
    else Rejected
        A-->>C: ❌ Notify: Campaign Rejected (with reason)
    else Rework Required
        A-->>C: 🔄 Notify: Rework Required (remarks)
        C->>A: Resubmit Updated Campaign
        A->>A: Re-review Campaign
    end
    end

    rect rgb(255, 245, 230)
    Note over A,V: PHASE 2 – Vendor Allocation & Site Survey
    A->>V: Allocate Vendor to Campaign
    Note over V: Vendor notified of assignment
    V->>V: Ground team visits site(s)
```

V-->V: Upload Billboard Images
(size, description, location/GPS,
media type, multiple photos)

V-->>A: Site Survey Complete (images + metadata)

Note over V: PDF Report auto-generated
from uploaded images
end

rect rgb(230, 255, 230)

Note over A,V: PHASE 3 – Image Selection & Creatives

A-->>A: Review Site Survey PDF

A-->>A: Select images (1 or many from survey)

A-->>A: Prepare Creatives for each selected image

Note over A: Creatives processing complete
end

rect rgb(255, 230, 245)

Note over A,V: PHASE 4 – Work Order & Installation

A-->>V: Create & Send Work Order
(linked to creatives + same vendor)

V-->>V: Start Installation

V-->>V: Upload installation images
(one per creative approved)

V-->>A: Installation Complete
(Installation Report auto-generated)
end

rect rgb(245, 245, 230)

Note over C,A: PHASE 5 – Client Verification

A-->>C: Send Installation Report

alt Accepted

C-->>A:  Installation Accepted

else Rejected

C-->>A:  Installation Rejected (remarks)

A-->>V:  Rework Installation

V-->>V: Redo installation & re-upload

V-->>A: Updated Installation Report

A-->>C: Resend Report

else Rework Required

C-->>A:  Rework Required (remarks)

A-->>V: Forward rework instructions

end

end

rect rgb(240, 230, 255)

Note over C,V: PHASE 6 – Invoicing & Closure

A-->>V: Notify: Installation Accepted

V-->>A: Send Invoice

A-->>C: Forward Invoice

alt Invoice Accepted

C-->>A:  Invoice Accepted

A-->>A: Close Work Order

A-->>V: Notify: Work Order Closed

else Invoice Rejected

C-->>A:  Invoice Rejected (remarks)

A-->>V: Notify: Update Invoice

V-->>A: Resend Updated Invoice

```
A-->C: Forward Updated Invoice
end
end
```

Phase Summary

Phase	Key Actions
1. Campaign Creation & Review	Client creates → Admin approves/rejects/rework
2. Vendor Allocation & Site Survey	Admin assigns vendor → Vendor visits sites, uploads geotagged billboard photos → PDF auto-generated
3. Image Selection & Creatives	Admin selects images from survey → Prepares creatives for each
4. Work Order & Installation	Admin creates work order → Same vendor installs → Uploads proof images → Report generated
5. Client Verification	Admin sends report → Client accepts/rejects/rework
6. Invoicing & Closure	Vendor invoices → Admin forwards → Client accepts/rejects → Admin closes work order

4. Campaign State Machine

17 states covering the full lifecycle:

```
stateDiagram-v2
    [*] --> pending_approval: Client creates campaign

    pending_approval --> approved: Admin approves
    pending_approval --> rejected: Admin rejects
    pending_approval --> rework_required: Admin requests rework

    rework_required --> pending_approval: Client resubmits

    approved --> vendor_allocated: Admin allocates vendor

    vendor_allocated --> survey_in_progress: Vendor starts site survey

    survey_in_progress --> survey_completed: Vendor uploads images + PDF generated

    survey_completed --> creatives_in_progress: Admin selects images & starts creatives

    creatives_in_progress --> creatives_ready: Admin finishes creatives

    creatives_ready --> work_order_issued: Admin creates work order

    work_order_issued --> installation_in_progress: Vendor starts installation
```

```

installation_in_progress --> installation_completed: Vendor uploads install
images + report

installation_completed --> client_verified: Client accepts installation
installation_completed --> client_disputed: Client rejects/requests rework

client_disputed --> installation_in_progress: Vendor reworks installation

client_verified --> invoiced: Vendor sends invoice via Admin

invoiced --> invoice_accepted: Client accepts invoice
invoiced --> invoice_rejected: Client rejects invoice

invoice_rejected --> invoiced: Vendor resends updated invoice

invoice_accepted --> closed: Admin closes work order

closed --> [*]
rejected --> [*]

```

Status Transitions Table

From	To	Triggered By	Action
—	pending_approval	Client	Create campaign
pending_approval	approved	Admin	Approve campaign
pending_approval	rejected	Admin	Reject campaign
pending_approval	rework_required	Admin	Request rework (with remarks)
rework_required	pending_approval	Client	Resubmit updated campaign
approved	vendor_allocated	Admin	Allocate vendor
vendor_allocated	survey_in_progress	Vendor	Start site survey
survey_in_progress	survey_completed	Vendor	Upload images (PDF auto-generated)
survey_completed	creatives_in_progress	Admin	Select images, begin creatives
creatives_in_progress	creatives_ready	Admin	Finish creatives
creatives_ready	work_order_issued	Admin	Create work order
work_order_issued	installation_in_progress	Vendor	Start installation
installation_in_progress	installation_completed	Vendor	Upload install images + report

installation_completed	client_verified	Client	Accept installation
installation_completed	client_disputed	Client	Reject/rework installation
client_disputed	installation_in_progress	Vendor	Redo installation
client_verified	invoiced	Vendor	Send invoice
invoiced	invoice_accepted	Client	Accept invoice
invoiced	invoice_rejected	Client	Reject invoice (with remarks)
invoice_rejected	invoiced	Vendor	Resend updated invoice
invoice_accepted	closed	Admin	Close work order

5. Data Model (ER Diagram)

```

erDiagram
    USER {
        ObjectId _id PK
        String name
        String email UK
        String password
        Enum role "admin | client | vendor"
        Date createdAt
    }

    CAMPAIGN {
        ObjectId _id PK
        String title
        String brief
        Number budget
        StringArray targetLocations
        Date startDate
        Date endDate
        Enum status "17 states"
        ObjectId clientId FK
        ObjectId vendorId FK
        String adminRemarks
        Date createdAt
        Date updatedAt
    }

    SITE_SURVEY {
        ObjectId _id PK
        ObjectId campaignId FK
        ObjectId vendorId FK
        Array images "embedded SurveyImage"
        String pdfUrl
        Enum status "pending | completed"
  
```

```

    Date createdAt
}

SURVEY_IMAGE {
    String imageUrl
    String description
    String size
    GeoJSON location "lat and lng"
    String locationAddress
    Enum mediaType "6 types"
    Boolean selectedByAdmin
}

CREATIVE {
    ObjectId _id PK
    ObjectId campaignId FK
    ObjectId surveyImageRef FK
    String creativeImageUrl
    String description
    Enum status "pending | processed"
    Date createdAt
}

WORK_ORDER {
    ObjectId _id PK
    ObjectId campaignId FK
    ObjectId vendorId FK
    Array creativeIds FK
    Enum status "issued | in_progress | completed | closed"
    Date issuedDate
    Date closedDate
}

INSTALLATION {
    ObjectId _id PK
    ObjectId workOrderId FK
    ObjectId campaignId FK
    ObjectId vendorId FK
    Array images "install proof"
    String reportPdfUrl
    Enum status "in_progress | completed | disputed | verified"
    String clientRemarks
    Date createdAt
}

INVOICE {
    ObjectId _id PK
    ObjectId campaignId FK
    ObjectId workOrderId FK
    ObjectId vendorId FK
    Number amount
    String invoiceFileUrl
}

```

```

    Enum status "pending | accepted | rejected"
    String clientRemarks
    Date createdAt
    Date updatedAt
  }

  NOTIFICATION {
    ObjectId _id PK
    ObjectId userId FK
    String title
    String message
    Enum type "14 event types"
    ObjectId referenceId
    String referenceModel
    Boolean read
    Date createdAt
  }

  USER ||--o{ CAMPAIGN : "creates (client)"
  USER ||--o{ CAMPAIGN : "assigned to (vendor)"
  CAMPAIGN ||--|| SITE_SURVEY : "has"
  SITE_SURVEY ||--o{ SURVEY_IMAGE : "contains"
  CAMPAIGN ||--o{ CREATIVE : "has"
  SURVEY_IMAGE ||--o{ CREATIVE : "selected for"
  CAMPAIGN ||--|| WORK_ORDER : "has"
  WORK_ORDER ||--o{ CREATIVE : "references"
  WORK_ORDER ||--|| INSTALLATION : "has"
  CAMPAIGN ||--|| INVOICE : "has"
  USER ||--o{ NOTIFICATION : "receives"

```

Media Types (for Survey Images)

Value	Display Name
vinyl	Vinyl
one_way	One Way
sunboard	Sunboard
no_lit_board	No-Lit Board
glow_sign_board	Glow Sign Board
acrylic_board	Acrylic Board

6. Notification Events

```

graph LR
  subgraph "Notification Events"
    N1["campaign_reviewed<br/>-> Client"]
  end

```



```

N2["campaign_rework<br/>→ Client"]
N3["vendor_allocated<br/>→ Vendor"]
N4["survey_completed<br/>→ Admin"]
N5["creatives_ready<br/>→ Admin internal"]
N6["work_order_issued<br/>→ Vendor"]
N7["installation_completed<br/>→ Admin"]
N8["installation_report_sent<br/>→ Client"]
N9["installation_verified<br/>→ Admin then Vendor"]
N10["installation_disputed<br/>→ Admin"]
N11["invoice_received<br/>→ Client"]
N12["invoice_accepted<br/>→ Admin"]
N13["invoice_rejected<br/>→ Admin then Vendor"]
N14["work_order_closed<br/>→ Vendor"]
end

subgraph "Socket.IO Rooms"
  R1["room: user_{userId}"]
  R2["room: role_admin"]
end

N1 & N2 --> R1
N3 & N6 & N14 --> R1
N4 & N7 & N10 & N12 & N13 --> R2
N8 & N11 --> R1
N9 --> R1

```

Notification Events Detail

#	Event	Recipient	Trigger
1	campaign_reviewed	Client	Admin approves or rejects campaign
2	campaign_rework	Client	Admin requests campaign rework
3	vendor_allocated	Vendor	Admin assigns vendor to campaign
4	survey_completed	Admin	Vendor completes site survey
5	creatives_ready	Admin (internal)	Creatives processing finished
6	work_order_issued	Vendor	Admin creates work order
7	installation_completed	Admin	Vendor finishes installation
8	installation_report_sent	Client	Admin forwards installation report
9	installation_verified	Admin → Vendor	Client accepts installation
10	installation_disputed	Admin	Client rejects installation
11	invoice_received	Client	Admin forwards invoice
12	invoice_accepted	Admin	Client accepts invoice
13	invoice_rejected	Admin → Vendor	Client rejects invoice

14	work_order_closed	Vendor	Admin closes work order
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7. API Endpoints

Resource	Method	Endpoint	Role	Description
Auth	POST	/api/auth/register	Public	Register user
Auth	POST	/api/auth/login	Public	Login, returns JWT
Campaign	POST	/api/campaigns	Client	Create new campaign
Campaign	GET	/api/campaigns	All	List campaigns (filtered by role)
Campaign	GET	/api/campaigns/:id	All	Get campaign details
Campaign	PUT	/api/campaigns/:id	Client	Update campaign (resubmit)
Campaign	PATCH	/api/campaigns/:id/review	Admin	Approve/reject/rework campaign
Campaign	PATCH	/api/campaigns/:id/allocate-vendor	Admin	Assign vendor
Site Survey	POST	/api/site-surveys	Vendor	Create survey with images
Site Survey	GET	/api/site-surveys/:campaignId	Admin, Vendor	Get survey for campaign
Site Survey	PATCH	/api/site-surveys/:id/select-images	Admin	Select images from survey
Creative	POST	/api/creatives	Admin	Create creative from selected image
Creative	GET	/api/creatives/:campaignId	Admin, Vendor	List creatives for campaign
Creative	PATCH	/api/creatives/:id	Admin	Update creative status
Work Order	POST	/api/work-orders	Admin	Create work order
Work Order	GET	/api/work-orders/:id	Admin, Vendor	Get work order details
Work Order	PATCH	/api/work-orders/:id/close	Admin	Close work order
Installation	POST	/api/installations	Vendor	Start installation + upload images
Installation	GET	/api/installations/:workOrderId	All	Get installation details

Installation	PATCH	/api/installations/:id/verify	Client	Accept/reject/rework installation
Invoice	POST	/api/invoices	Vendor	Create invoice
Invoice	GET	/api/invoices/:campaignId	All	Get invoice for campaign
Invoice	PATCH	/api/invoices/:id/review	Client	Accept/reject invoice
Invoice	PUT	/api/invoices/:id	Vendor	Update rejected invoice
Notification	GET	/api/notifications	All	Get user's notifications
Notification	PATCH	/api/notifications/:id/read	All	Mark notification as read

8. Tech Stack

Layer	Technology
Frontend	React 18 + Vite
UI Components	Tailwind CSS / Material UI
State Management	React Context / Zustand
Backend	Node.js + Express
Database	MongoDB + Mongoose
Authentication	JWT (JSON Web Tokens)
Real-Time	Socket.IO
File Uploads	Multer (local) / Cloud Storage
PDF Generation	PDFKit / Puppeteer
Maps & GPS	Google Maps JavaScript API
Camera	HTML5 MediaDevices API